LUXNA RAMNAUTH

Junior Data and AI Engineer
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SUMMARY

Junior Data and AI Engineer and Software Engineering graduate who builds ETL/ELT pipelines, SQL data models and data science and machine learning solutions, from raw data collection through to model results. Currently building a RAG-based banking chatbot in PyTorch and Hugging Face and an AI-powered fraud detection pipeline. Across three internships, cleaned and validated large datasets, helped reduce post-release defects by 5%, and contributed to a projected 15% gain in operational efficiency. Works well in international teams spanning Mauritius, Vietnam and Switzerland, and explains model behaviour clearly to non-technical stakeholders.

CORE SKILLS

- **Data Engineering:** ETL/ELT data pipelines (collection, cleansing, transformation, storage); data models, data warehouses, data lakes; data validation and quality checks; API integration
- **Databases and SQL:** SQL (MS SQL Server, MariaDB, Oracle SQL); database systems and administration; query optimisation; NoSQL exposure
- **AI and Machine Learning:** Deep Learning; Generative AI and LLM applications; Natural Language Processing; RAG architecture; PyTorch; TensorFlow; Hugging Face; scikit-learn; regression; time series forecasting; anomaly detection; model evaluation; feature engineering; Data Science
- **Software Engineering and Programming:** Software engineering practices (Git, Agile, API integration); Python (Pandas, NumPy, scikit-learn, PyTorch, Matplotlib); SQL; R; Java; Git; Jupyter Notebook
- **Cloud and Data Tools:** AWS, GCP, Azure (Azure Data Factory), Snowflake, Databricks, Apache Spark (declarative pipelines)
- **Data Visualisation and BI:** Power BI; Tableau; Matplotlib; dashboard design; data-driven decision support
- **Responsible AI and Governance:** Generative AI application evaluation and governance; data quality, validation and regulatory compliance monitoring
- **Testing and Delivery:** Regression, functional, performance and API testing; Agile and sprint-based delivery
- **Soft Skills:** Cross-functional collaboration; clear communication with technical and non-technical stakeholders; adaptability; continuous learning

AI AND DATA PROJECTS

RAG-Based Banking Chatbot (in development, since July 2026)

- Retrieval-augmented generation chatbot for banking use cases, applying natural language processing with a custom pipeline in PyTorch and Hugging Face.
- Covers document ingestion, embedding generation, retrieval, and integration of retrieved context with the language model for response generation.

Fraud Detection Pipeline with AI-Powered Analysis (in development, since July 2026)

- End-to-end machine learning pipeline for anomaly and fraud detection.

Bank Customer Churn Prediction (June 2026)

- Predictive machine learning model for customer retention.

Final Year Project: Fashion Assistant Mobile Application (2025)

- Solo, full-stack, AI-powered mobile application built with TensorFlow.

WORK EXPERIENCE

Information System Analyst Intern, Central Electricity Board (CEB), Curepipe, Mauritius | Dec 2023 – Jan 2024

- Built end-to-end ETL data pipelines covering collection, cleansing, preprocessing, model training and results presentation.
- Applied scikit-learn regression models to five years of historical consumption data.
- Acquired, processed and cleansed nine structured, large-scale datasets in Jupyter Notebook; defined collection and validation processes for downstream analysis.
- Used statistical metrics and pattern and anomaly detection to surface insights, contributing to a projected 15% improvement in operational efficiency.

- Created trend and seasonal reports in Matplotlib, translating findings into decision-ready insights for management.

Associate QA Engineer Intern, ELCA Group, St Pierre, Moka, Mauritius | Sep 2025 – Apr 2026

- Executed 30+ structured test cases (regression, functional, performance and API) within an Agile framework across three international client projects, reducing post-release defect rates by 5%.
- Analysed sprint-level defect and execution data to identify patterns, supporting debugging and optimisation and delivering actionable recommendations on data quality and governance.
- Collaborated daily in a nine-person cross-functional team spanning Mauritius, Vietnam and Switzerland in a fast-paced, deadline-driven environment.

Associate Software Engineer Intern, Agileum Ltd, Floral, Vacoas, Mauritius | Mar 2024 – Apr 2024

- Optimised 20+ SQL queries and API-driven backend reporting to ensure data accuracy and consistency across multiple sources.
- Delivered feature enhancements and defect resolution across two concurrent client projects, coordinating with project managers and business teams.

EDUCATION

BSc Applied Computing, University of Mauritius, Réduit, Moka | August 2025 (08/25)
Relevant coursework: Database Systems and Administration; Programming and Data Structures; Mathematics for Computing; High Performance Computing; Software Modelling; Applied Cybersecurity; Mobile Computing and Wireless Technologies; Business Computing

Higher School Certificate (A-Level), Swami Sivananda State Secondary School | 2021
Main subjects: Computer Science, Business Studies, Economics, Accounting. Sub-subjects: General Paper, Sociology

School Certificate, Swami Sivananda State Secondary School | 2017

CERTIFICATIONS

- Google IT Automation with Python Professional Certificate (Google, Coursera)
- GitHub Foundations (GitHub)
- Microsoft Generative AI Engineering Professional Certificate (Microsoft)
- Microsoft AI and Machine Learning Engineering Professional Certificate (Microsoft)
- IBM RAG and Agentic AI Professional Certificate (IBM)
- IBM AI Engineering Professional Certificate (IBM)
- IBM Data Engineering Professional Certificate (IBM)
- Generative AI for Data Engineers Specialization (IBM)
- Generative AI Application Evaluation and Governance (Databricks)
- Build Data Pipelines with Apache Spark Declarative Pipelines (Databricks)
- Databricks Performance Optimization (Databricks)
- Machine Learning Specialisation Certificate (DeepLearning.AI)
- Microsoft Azure Fundamentals, AZ-900 (DataCamp)
- Google Advanced Data Analytics Professional Certificate (Google)
- Google Data Analytics (Coursera)
- Google Business Intelligence (Coursera)
- Google AI Essentials (Coursera)
- Business Finance and Data Analysis Fundamentals Specialization (Rice University, Coursera)
- Deloitte Australia: Data Analytics Job Simulation (Forage)

LANGUAGES

English: Proficient | French: Proficient | Hindi: Elementary | Marathi: Elementary

REFERENCES

Available upon request.